<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>6</td><td>(a)</td><td></td><td>Attempt at starting from Faraday's law e.g. $\frac{BA}{t}$ (1)
Clear explanation leading to Bvd – minimum $\frac{BA}{t} = \frac{Bxd}{t} =$ Bdv OR area in 1 s is dv (1)</td><td>2</td><td></td><td></td><td>2</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>Flux in or right pd (in 1 s) = B_2vd (1)
Flux out or left pd (in 1 s) = B_1vd (1)
Net flux change or pd around the loop (per s) = B_2vd - B_1vd (1)
Alternative:
$V = \frac{d}{dt}(BA) = \frac{AdB}{dt}$ (1)
$V = \frac{AdB}{dx}\frac{dx}{dt}$ (1)
$V = d^2\left(\frac{B_2 - B_1}{d}\right)v$ QED (1)
Award 2 marks for:
pd on right = B_2vd
pd on left = B_1vd
Hence B_2vd - B_1vd
Award 1 mark for:
It depends on the change in flux / B
Hence ΔBvd</td><td>1</td><td>1
1</td><td></td><td>3</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>B_2 - B_1 = 1.20 × 0.038 (1)
Answer = 15.25 m[V] or 1.2 × 0.038 × 8.8 × 0.038 seen (1)</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td></td></tr>
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Question			Marking details	Marks available					
				AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
		(iii)	Length = 4 × 3.8cm (1) Area = 2.2 mm × 2 mm (1) Resistivity equation used (0.55 mΩ) (1) $I = \frac{V}{R}$ used (1) Correct answer = 28 [A] (1) **ecf** 56 A and 111 A award 4 marks	 1 1	1 1 1		5	3	
			Question 6 total	**5**	**7**	**0**	**12**	**7**	**0**